

Augmented Foot: A Comprehensive Survey of Augmented Foot Interfaces

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ABSTRACT

Augmented foot interfaces have been studied since the beginning of wearable computers. The worlds' first wearable computer was an instrumented shoe that consisted of a toe operated switch with a wireless module. Since then, academic research and commercial products on augmented foot interfaces are booming with novel interfaces every year. This paper surveys the body of work on augmented foot interfaces and shows the current trends and guidelines for future augmented foot interfaces. We contribute a classification of over 100 academic papers and commercially-available products. We discuss the integration of augmented foot interfaces, interaction schemes and application domains. Finally, we contribute a set of design considerations to scaffold future research of augmented foot interfaces based on the classification and inspired by the surveyed work.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing.

KEYWORDS

Foot Augmentation, Human Augmentation, Foot Interactions, Survey, Augmented Human, Augmented Foot

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1 INTRODUCTION

Research in Human-Computer Interaction and the rise of wearable computing enabled advances in human augmentation. Prior research and commercial products showcased wearable computers for different body locations in different form-factors. Reportedly, the first wearable computer was developed for the foot by Edward

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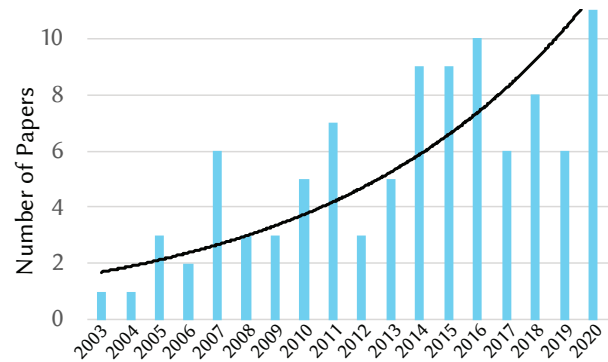


Figure 1: Amount of publications on AFIs from 2003 until today.

and Shannon [112] to predict the outcome of casino roulette wheels inconspicuously by using a toe switch in their shoes. In this scenario, both the ease of integration with the human body and discreetness can be considered as the main reasons for selecting the foot as an interaction device over other body parts.

Most of the early foot interfaces were not wearable. They were designed as stationary input devices for hands-free interactions. For example, before Engelbart et al. [24, 25] introduced the mouse, they explored different foot-operated input interfaces. Since then, there was a steady stream of publications pertaining to foot interfaces. Velleo et al. [114] surveyed this large body of work in 2015. They identified three main categories of foot interfaces: foot-operated devices, foot-tracking and foot-worn devices.

Over the past decade, advances in micro-electronics and Ubiquitous Computing have transformed such wearable devices into human augmentation devices: they seamlessly integrate with a users' mind, body, and behavior [73]. This observation was subsumed in a sub-field of human augmentation dubbed Assistive Augmentation [42].

Both trends, foot interfaces and Assistive Augmentation, have clear overlaps, which we identify in this paper. We call interfaces at this very intersection augmented foot interfaces (AFI). We define AFIs as *self-contained foot-worn mobile interfaces, with sensing or output capabilities, that extend human abilities*. AFIs mainly enhance human capabilities in three ways: (1) sensing physical parameters of the body such as movement and pressure (input), (2) providing feedback to the wearer (output) and (3) using a combination of sensing and feedback. The end effect is not necessarily focused

on the foot itself; rather, it may be on another part of the body or the entire body. For example, CabBoots [31] augment the foot by providing tactile feedback to the foot to scaffold user navigation. In GymSoles [21, 22], foot augmentation improves the overall body posture while StressFoot [23] improves cognitive awareness.

In this paper, we tackle the overarching research questions of (a) what are sweet spots for AFI research and (b) what are common themes of existing AFI research, e.g., in terms of devices used, interaction schemes employed, and applications targeted. While prior art surveyed the general research avenue of foot interfaces, unraveling the notion of assistive augmentation for foot interfaces requires a more zoomed-in view. We aim to fill this gap. Our work contributes to the Augmented Humans community by mapping out the research landscape of augmented foot interfaces. To this end, our contributions are as follows:

- **Review and classification of Augmented Foot Interfaces:**

We provide a comprehensive literature review of work at the intersection of Assistive Augmentations and foot interfaces and map out the research landscape of AFIs. We synthesize review results into a classification of augmented foot devices.

- **Design Considerations and Guidelines:** Based on our theoretical classification, we identify unique opportunities for AFIs for human augmentation. We translate these opportunities into a set of design considerations for future research on AFIs.

2 CLASSIFICATION OF AUGMENTED FOOT INTERFACES

In the following, we survey 112 academic publications from journals, conference proceedings and 9 commercial devices. All the papers were cited in relevant sections of the paper. Hence, the list of papers that we reviewed are included in the references section. The papers were selected using keyword search methods through academic publication engines (google scholar, ACM Digital Library, IEEE Explore), hierarchical citation-tree backtracking from existing publications, and general queries in a web search engine. The keywords we used are: ‘Smart Insoles’, ‘Foot Pressure’, ‘Foot Interaction’, ‘Foot Augmentation’, ‘Gait Analysis’, ‘Center of Pressure’, ‘CoP’, ‘Gait Augmentation’, ‘Balance’, ‘Foot Pressure Sensing’, ‘Motion Analysis’.

Before discussing the respective papers, we detail our approach. In our classification, we followed a hybrid inductive and derivative approach. We first used an inductive approach to define the main classes, meaning we defined classes at the beginning and tagged papers under relevant classes. These classes were mainly inspired by prior work [46, 103, 114]. Later, we used a derivative approach to identify further sub-classes. Below we further describe the classification approach in detail.

Our classification considers 5 dimensions: form factor, input modalities, output modalities, application domains, and interaction schemes (See Figure 2). The first four dimensions were inspired by prior work on finger augmentation devices [103]. The interaction scheme dimension was adapted from prior surveys on foot [114] and hand interactions [46]. From the above mentioned classes, input and output modalities, and form factors discuss the physical

affordances of AFIs. It describes how different sensors and actuators interfaced or integrated with the user’s foot sense/provide an input/output. The sub-classes of form factors and input and output modalities were further derived based on our surveyed work. In interaction schemes, we classify 2 categories of actions that users employ when interacting with AFI using their feet: (1) implicit interactions and (2) explicit interactions. Prior work has clustered explicit interactions into three categories: Semaphoric, Deictic, and Manipulative [46, 114]. Semaphoric actions are mainly referred to as specific foot gestures, while in manipulative interactions, foot movements or pressure is used to control an external device. However, AFIs could not be categorized under deictic as deictic devices always need an external target or tracking device to complete the interaction. Hence, we only divided Explicit interactions into 2 subcategories semaphoric and manipulative. Finally, we discuss the rich application domains of AFIs. We derived several application domains that clearly categorize what capabilities are enhanced or what type of problems are solved by AFIs.

2.1 Form Factor

To support different interactions and integrate sensors and actuators to the foot, AFIs have been designed in different ways. They can be attached to the foot in several wearable form factors in the different anatomical location of the foot.

Many form factors are designed in the form of daily foot worn accessories such as shoes [85, 88], sandals [55, 84], socks [8, 16], and insoles [119, 120] (See Figure 3). Some are designed as bands or tags which can attach to the ankle [34, 92, 127], leg [58, 105] or thigh [127]. Instead of designs conforming to a typical foot-worn accessory, some form factors only target a specific part of the foot, such as the big toe [94] or the ankle [69]. Some form factors are designed as sensor tags that can mount on the shoe [5, 6, 18]. Some interfaces only use one form factor, while some interfaces may use both form factors at once. For example, Sensoria [102] is a pressure sensor that is embedded into the sock with a sensor tag attached to the ankle. The sensor consists of an IMU sensor and a battery with data storage. StressFoot [23] has a smart insole connected to an ankle strap which consists of an IMU.

Below is the summary of the form factors that we identified from the survey. The instances which have 2 form-factors are included in both categories.

- Insoles (65 instances)
- Shoes (34 instances)
- Sensor tags/ Ankle Straps (15 instances)
- Sandal/Slippers (4 instances)
- Socks (5 instances)

The most frequent form factors evident in literature are insoles; this could be because an insole can be worn inside any shoe type if the size is well-matched. Also, insoles are unobtrusive and not highly noticeable to the outside world. Therefore, applications that require unobtrusiveness have mainly used form factors such as insoles and shoes. One good example is Edward’s wearable computer hidden inside a shoe helping to predict the outcome of a roulette inconspicuously [112]. Many other works have also tracked physical parameters unobtrusively using a shoe [15] or an insole [19]. Similarly, the sock is another unobtrusive form factor used for subtle

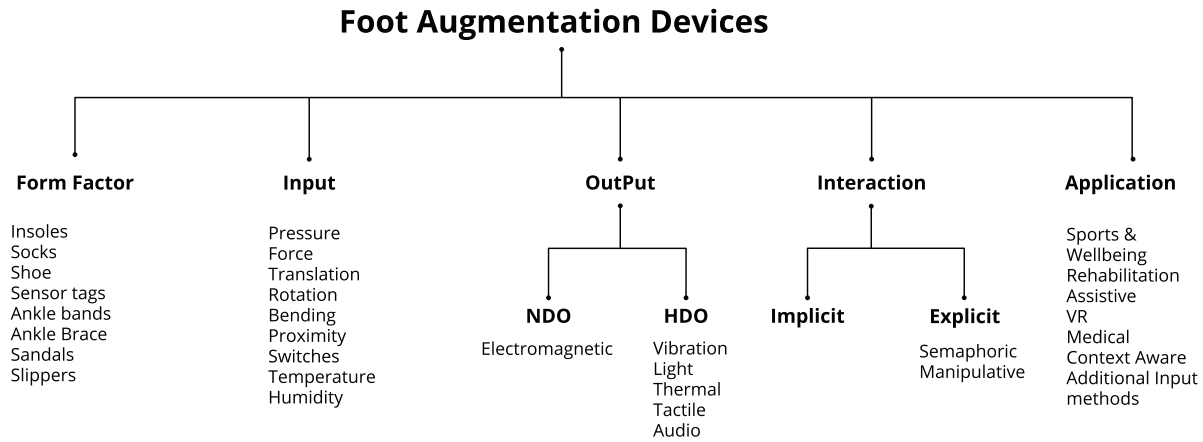


Figure 2: Classification hierarchy of AFIs

foot gestures. [32]. In most applications where the foot was used as a place to provide feedback, a sock has been used to embed vibration motors [76]. Sandals or slippers are another form factor that some prior work has used. These works used sandals or slippers as a form factor for VR locomotion applications [54, 108]. Since these devices are meant to be shared with others, the convenience of sandals or slippers may be the main reason for using such a form factor.

Instead of permanently embedding sensors to a typical wearable accessory, some works have used temporary augmentation by temporarily attaching a sensor to a shoe or a sock [5, 18]. The main advantage of such attachment is augmenting any foot-wear by using a single sensor tag. Some of these sensor tags are also designed to add aesthetic value to the foot-wear [18]. Also, sensor tags have been attached to certain foot locations, such as the ankle [3, 127]. Most of these sensor tags contain sensors like IMUs that track the lower limbs' motion and infer joint angles, such as the knee angle or ankle angle [23].

2.2 Input Modalities

Most of our surveyed work reported at least one kind of input modality. We identified the most commonly used modalities from our classification, either sensing a change of condition of the foot (e.g. humidity, temperature etc) or making the foot to explicitly change a parameter (bending, rotation, pressing on a switch). Furthermore, we discussed underlying technologies used to create the input modality. For example, foot pressure or impact force is an input modality that can sense using different technologies such as force-sensitive resistors, Capacitive sensors, or piezo-resistive sensors.

Below is the summary of input modalities that we identified from the survey. Note that some instances may have more than one input modality.

- Pressure or force between foot and contact surface (94 instances)
- Translation (44 instances)
- Rotation (32 instances)

- Bending (11 instances)
- Proximity to other devices or limbs (7 instances)
- Binary state buttons (4 instances)
- Temperature (3 instances)
- Humidity (1 instance)

Pressure sensors are widely used in AFIs since it easily integrates with the foot as a pressure-sensitive insole [9, 83, 107] or a smart shoe [43, 90] with embedded pressure sensors. Users with different foot sizes may be the main disadvantage of sensing pressure, as relative sensor placements of an insole changes according to the foot size. The primary usage of these pressure sensors is to identify the pressure exerted on the foot to infer phases of the gait cycle and physical activities such as walking, running, standing, and sitting [59]. In VR applications, pressure sensors were used as an input for VR locomotion [17, 111]. For example, users can lean forward or backward to move forward or backward in the VR world by changing the fore-foot and rear foot pressure [17]. In some work, pressure sensors embedded into smart insoles/shoes were used to implement walk-in-place VR locomotion [12, 111]. To measure pressure, researchers have used different technologies such as Force Sensitive Resistors (FSR) [2, 57, 79, 93], Capacitive Sensors [11, 63, 67, 120], Piezo-electric sensors [36, 38, 61], Strain Guages [50, 68], polyvinylidene fluoride (PVDF) sensors [86], ElectroMechanical Film (EMFi) sensors [37], Air pressure sensors [51, 52], and Conductive Polymer sensors [119, 128]. In many works, multiple pressure sensors were embedded to the insole to identify the pressure of different areas of the plantar foot such as rear, mid, and front foot [63, 121].

The next most popular input modalities are translation and rotation, which primarily uses Inertial Measurement Units for sensing. IMUs are implemented using accelerometers (to sense motion) and gyroscopes (to sense orientation). Similarly to pressure, translation and rotation modalities have been mostly used to infer gait phases [118], activity recognition [78, 122], or to identify foot motion [40]. In literature, multiple IMUs have been used to identify the foot's joint angles, such as knee angle and ankle angle [20, 30]. However, when using multiple IMUs to detect angles, proper sensor calibration and placement is necessary. Also, similar to pressure

sensors, IMUs have been used to implement walk-in-place VR locomotion [80, 118]. In contrast to pressure, one of the main advantage of sensing translation using IMUs is the ability to directly identify the direction the foot moves. This has been used for indoor localization in prior applications [56]. Sensing bend is another modality used in many applications that infer joint angles, such as knee joints and [81] ankle joints [87, 89]. Measurement of the joint angle has been used in applications for detecting gait for rehabilitation [85] and to measure exercise performance [7]. The main advantage of sensing bend is the easier integration of sensors with textiles. Hence, on many occasions, bend sensors seemed to be embedded in knee braces measuring knee angles. Also, in some work, ankle bend and insole bend [11, 15, 41] have been measured by using bend sensors.

Binary state button is another modality that has been used. These state buttons were mainly controlled using natural gestures, such as toe or heel taps. One example is the smart shoe, which uses a toe operated switch to convey roulette results [112]. Another example used flexible switches embedded in an insole to control robotic limbs [13]. Also, FSR switches [94] and micro switches [126] have been used to track tapping gestures.

There were some occasions that prior work used distance as an input modality. For example, CabBoots [31] has used a proximity sensor to identify uneven terrains for a navigation system. Also, prior work has used sonar sensors and electric field sensors integrated into shoes to identify the distance from the ground to the shoe for interactive footwear. Use of such sensors are mainly for dancing and therapy [11, 87, 89, 90].

In AFIs designed for diabetic ulcer prevention, temperature [53, 70] sensors and humidity [70] sensors are frequently used, in addition to pressure sensors. These temperature and humidity sensors track the thermal profile of the foot to infer ulceration possibilities. Usually, high temperature differences between the feet's identical location and the plantar's dry skin mean an individual is at a high risk of getting foot ulcers. Also, a commercially available insole, Digitsole [19], senses the temperature of the foot to keep the foot warm during cycling or running activities.

2.3 Output Modalities

Output in AFIs is mostly generated towards the user. It may provide real-time feedback on a user's performed action or provide information from an external system. We considered two classes of outputs, ones with Human-Detectable Output (HDO) and ones with Non-Detectable Output (NDO), introduced by Shilkrot et al. [103]. These classes are also suitable in the AFI context. HDOs mainly generate energy detectable by the innate human senses, while NDOs generate non-detectable energy, such as electromagnetic fields. Although AFIs with HDOs are considered widely explored, most works had NDOs (HDO – 27 instances, NDO 88 instances). From 27 HDO instances, only three were purely output AFIs. Interestingly, those devices were designed either for navigation [45, 66] or non-verbal communications [39]. Most of the HDOs convert an information extract from an input to a human-understandable format.

The most common HDO modality is vibration, which has been used in 16 instances. The foot's high density of mechanoreceptors make it sensitive to vibration, rendering the foot as one of the best body locations to receive information as vibration patterns. For

example, GymSoles [22] and GymSoles++ [21] provide the center of pressure (CoP) feedback as a vibration for the foot. Similarly, work like climbing assist [27] provides vibration feedback about the intensity of pressure exerted at climbing shoes. Also, a commercially available product, Walkasins [116], provides foot pressure information to the diabetic neuropathy patients' ankle. Rhythm and shoes [84] understand the speed of the gait cycle and provides vibration feedback to change it. Some work has also used rhythmic vibration feedback on the foot to induce walking cycle speed or skill transfer [117]. In VR related applications, vibration has been widely used to recreate ground surface sensations [108], walking stairs [74, 75] or other sensations such as tickling [54]. The easy controllability and ability to render rich feedback could be the reason for the popularity of vibration as an HDO modality for AFIs.

Besides vibration output, some work changes footwear's morphology, generating tactile output according to impact force. For instance, CabBoots [31] changes the shoe surface's angle by using servo motors to provide feedback about navigation direction. Similarly, changing the foot sole's inclination is another HDO modality that has been used in AFIs designed for rehabilitation [6]. Pneumatic actuators are another type of method used to provide tactile output by redistributing foot pressure [57].

Visual or light feedback is another HDO modality that has been predominantly used in AFIs with a knee brace form factor. RGB Neo-pixel [81] strips or Electroluminescent [7] displays were used to provide feedback about correct knee angle during exercises such as squats and deadlifts. This visual output modality was mainly used to convey information to an external party such as a trainer. In addition to the above HDOs, Peltier elements were used to provide thermal as an output modality [63]. These elements are mainly used to recreate the sensations of different ground surfaces with other output modalities like vibration or tactile. Some commercial devices only used thermal to keep the foot temperature high solely to comfort the user [19].

NDO AFIs are mainly used to communicate information to other devices, mostly to a companion mobile device or a PC for post-analysis of data [52, 104, 127]. Electromagnetic waves such as Bluetooth [104], radio, and WiFi are the most commonly used NDOs. Some NDO AFIs send information to manipulate external devices such as external limbs [97]. Other NDOs send information to devices that provide feedback to another body part, such as head-mounted displays [21, 80] or headphones directly [40, 65, 82] or via companion mobile device [78].

The following is the summary of the most frequent output modalities that we encountered in the surveyed work.

- NDO Electromagnetic (88 instance)
- HDO Vibration (16 instances)
- HDO Light (4 instances)
- HDO Tactile (Other than Vibration, e.g compression) (6 instances)
- HDO Thermal (2 instances)
- HDO Audio (3 instances)

2.4 Interaction Scheme

In this subsection, we discuss the interaction between the AFIs and the user. We mainly discuss 2 classes of interactions: Explicit



Figure 3: Example form factors from prior work. A) Insole with an ankle strap [21], B) Insole from F-Scan [110], C) Socks with sensor tag from Sensoria [102], D) Intelligent Shoes for human identification. [41], E) Sandals for generating virtual material [108]. All the images are courtesy of their respective authors.

interactions and Implicit interactions. Explicit interactions can be further divided into two categories, as identified in the prior work of hand [46] and foot interactions [114]: semaphoric and manipulative. Implicit interactions cover actions that are not primarily aimed at interacting with a computerized system [100], such as gait identification. Interestingly, the majority of our surveyed work are devices that support implicit interactions. From devices for explicit interactions, 13 instances were semaphoric, while 4 instances were Manipulative.

- Semaphoric (13 instance)
- Manipulative (4 instances)
- Implicit (98 instances)

2.4.1 Explicit Interactions. Semaphoric actions can be considered as the main explicit interaction category that AFIs used. They are specific gestures intended to execute certain actions. The most common gesture found in literature is toe-tapping or heel tapping [109, 112, 126]. Some of these devices were used as an input for computer interfaces, especially for people with upper extremity disabilities [109]. Prior work has also explored a set of gestures that can be used using sock-placed pressure sensors [32]. Some of the foot-based semaphoric actions were used as an additional input modality combining with gaze input [94, 94, 95]. In addition to stationary gestures, AFIs were designed to implement gestures while walking, running, or standing for hands-free interactions [99, 105, 126].

The next popular application for semaphoric actions is locomotion in VR. Some applications defined several foot gestures [17, 115], while walking-in-place techniques directly used the walking action as input for locomotion [12, 62, 63, 111]. Some of those applications are even designed for rehabilitation [80]. In many of those applications, insoles with pressure sensors have been used to generate an input signal for the interaction.

Manipulative actions map foot movements and other parameters that the foot changes to external systems or devices. For example, controlling prosthetic hands by changing the pressure of a pressure-sensitive insole is a manipulative interaction [14]. Here, foot pressure parameters like intensity and spatial distribution are mapped to the prosthetic hand's control parameters. Another example is MetaArms [97, 98], where foot movements directly manipulate the wearable robotic arms. In this work, a foot-worn IMU sensor tag and an insole captures the movement and maps with the robotic limbs' movements. In another work, a AFI-based solution was explored to manipulate GUI elements by using foot pressure variations [121].

2.4.2 Implicit Interactions. According to Schmidt [100], in explicit interaction, the user tells the computer in a certain level of abstraction what they expect the computer to do, whereas in implicit interactions, the user's actions are not primarily aimed to interact with a computerized system, but the system understands those as input. The majority of work (95 instances) we surveyed fall under this category.

Most prior work in implicit interaction takes the user's gait movement (translation and rotation), foot pressure, or both, as input and use it in different applications. For example, gait parameters such as translation and rotation are used to identify abnormal gait patterns [15, 79], provide feedback to avoid injuries [22, 50], improve performance in sports and rehabilitation [26, 28, 72, 110], user identification [41, 64], user localization [31, 56], and activity recognition [16, 43, 120, 124]. The work that takes input as foot pressure uses that information for posture identification [62, 63, 83], identification of CoP to improve balance or posture [21, 22], activity recognition [16], identification of cognitive state [23], and terrain identification [64].

In some medical applications, AFIs were used to sense pressure [119, 128], humidity, and temperature [70] implicitly to provide feedback on the possibility of getting diabetic ulcers.

2.5 Applications

As we discussed in the earlier sections, AFIs use various input and output modalities to sense information or provide feedback. This has enabled various applications that improve or enhance human abilities (See Figure 4). Although AFIs were used as an additional input device for hands-free interactions in the early days, current trends mostly tend towards applications based on implicit interactions. However, in virtual reality, there is a rise in using AFIs as an input and output device to develop natural locomotion techniques and recreate natural sensations in a virtual environment. This section describes the major and minor application categories that we identified from the surveyed body of work.

The following are the major application domains we recognize for AFIs:

- Sports and Well-being (47 instances)
- Rehabilitation (14 instances)
- Assistive Devices (15 instances)
- VR Applications (14 instances)
- Medical Application (13 instances)
- Context Awareness (5 instances)
- mouse or keyboard like input (5 instances)

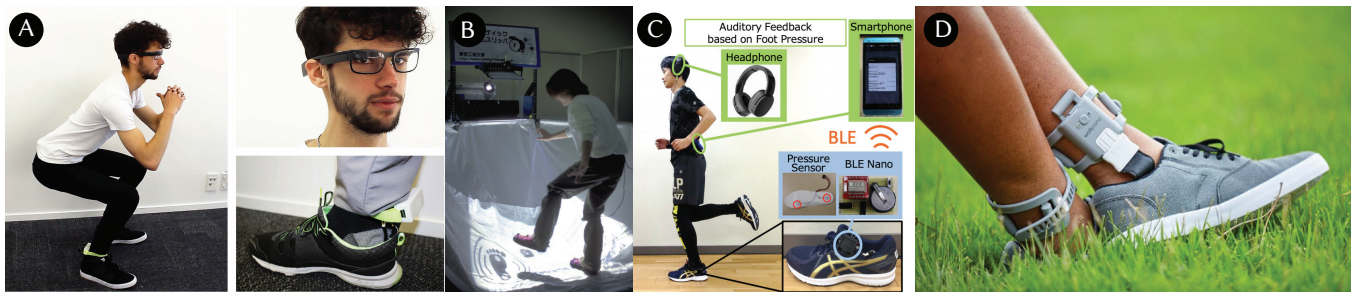


Figure 4: Example Applications of prior works. A) GymSoles++ [21] which improves exercise posture, B) Fantastic phantom slipper creates tactile feedback to recreate different sensations. [54], C) Improving balance of runners using auditory feedback [82], D) Walkasins' sensory neuroprosthesis to improve balance and mobility for people with sensory peripheral neuropathy.

2.5.1 Sports and Well-being. In our surveyed work, many of them are under the category of sports and well-being. From these instances, most of them are applications that enhance awareness on gait parameters or foot pressure to detect physical anomalies [15, 68]. Some AFIs enable personal tracking [72, 107], recognizing different activities [58, 122], intensity, and duration [28]. Also, there are novel applications that propose detecting stress levels using AFIs [23]. All these devices enhance the physical and mental awareness of a user for improved health and well-being.

Also, some applications improve gait while walking/running or enhance posture [60, 127] while exercising [7, 22, 81]. In these applications, some AFIs provide real-time feedback to improve those physical parameters. For example, GymSoles [22] and GymSoles++ [21] provide feedback about center-of-pressure directly to the foot to improve postures while performing squats and dead-lifts. Also, in another work, vibration feedback was provided to induce the gait cycle to increase or reduce walking speed [117]. Instead of providing direct feedback to the foot, such as tactile or vibration, some AFIs transferred information to an external device to provide auditory or visual feedback. These types of devices synced walking, running, or exercising with music and enhanced physical activities [78]. Some devices send data to a mobile application to post-analyze their gait and foot pressure parameters.

Also, in sports, AFIs were used to improve performance and skills [93]. For example, an IMU-based AFI was used to assess leg movements in tennis [3]. To enhance dancing skills, a shoe with an IMU and a sock with vibration motors was used for skill transfer in dance training [76]. A series of works was conducted by Paradiso et al. [84, 87, 89] to improve dancing and athlete performance based on a shoe-based AFI. Also, to enhance speed skaters' performance, a AFI was used, namely an IMU-based type of shoe [106]. Similar to some previous work, they used sonification as feedback to improve speed skating. ClimbAssist [27] is another work that used an insole and a shoe AFI to enhance climbing by providing real-time vibrotactile feedback about exerted pressure on the foot. In another AFI, a motorized shoe was used to adjust sole inclination according to the foot movements to prevent falling during rehabilitation [6].

2.5.2 Rehabilitation. Rehabilitation is another common application area that AFIs have been utilized, particularly for lower extremity rehabilitation. Many of those applications use shoe [90]

or insole [1, 2, 61] type AFIs embedded with pressure sensors and IMU sensor tags. Some of them are developed as socks [8] or knee braces [7] for easier wearability. Most of the rehab devices measure gait parameters and balance to provide post-analysis data about rehabilitation progress [44, 50, 92]. Some applications even used gamification techniques, such as VR games, to motivate the performance of rehab exercises [80]. Some AFIs provide rhythmic feedback to support walking rehabilitation [90]. In several applications, real-time feedback was used to inform participants whether they achieved the expected degree of motion. For example, Pt Viz [7] visualizes the knee angle extension by an electro-luminescent display attached to the knee brace sides.

2.5.3 Assistive Devices. AFIs have been identified in several assistive applications for both healthy users and users with disabilities. Mainly, they have been designed for controlling the prosthetic hands of people with disabilities [13]. Some work has also explored how AFIs like smart insoles or shoes with sensor tags can be used to control extra limbs [97, 98]. Moreover, a shoe-based keyboard has been developed to assist upper limb disabled users with typing [109]. Some work has also explored how AFIs can be used as assistive devices for hands-free interactions [125], especially to provide input [32, 121]. Several works have suggested AFI base implementations for control mobile devices while jogging or running [105, 123].

Also, AFIs have been used to assist the navigation of blind users and healthy users [31, 66]. In these applications, insoles or shoes use haptic feedback to provide navigation cues. In addition to communicating directions for navigation, haptic insoles have been used as an assistive device for language communication [39]. The foot's high sensitivity to haptics and unobtrusiveness makes it an ideal location to provide haptic feedback information. Also, in B-Foot [10], and AFI has been used to prevent falls in older adults. It senses the CoP of the users and uses small wheels embedded in the shoes to shift the foot backward to regain balance before falling.

2.5.4 VR applications. In virtual reality, AFIs have mainly been used to implement locomotion techniques or recreate sensations to enhance the immersive experience [17, 49, 96]. One popular VR related application for AFI is enabling walk-in-place locomotion [12, 63, 111]. In this locomotion technique, users perform stationary

walking action. An instrumented insole, shoe, or a foot-mounted sensor tag can capture this motion and map it to locomotion in VR [115]. Recent work has also used foot-based gestures for VR teleportation, one of the locomotion techniques used in VR.

The next popular AFI application for VR is recreating sensations for the foot, such as tickling [54] or stairs [74, 75], especially recreating different ground surface conditions by using AFIs. For example, BareFoot recreates virtual materials that can sense by foot using vibrotactile shoes [108]. ShoeSoleSense uses Peltier elements and vibration motors to change the shoe's temperature and texture to recreate different ground properties and certain environmental conditions inside the VR experience [62, 63].

2.5.5 Medical Applications. In the medical domain, AFIs have been designed for diagnosis, prevention, and providing treatments. Many devices have been designed to diagnose the possibility of getting diabetic foot ulcers (DFU). Many of them capture the foot pressure profile using a pressure-sensitive insole and provide a post-report about the possibility of getting DFUs [53, 91, 119, 128]. According to this information, patients can change their footwear. Some AFIs even have advanced capabilities measuring temperature, humidity, and pressure, which can better understand DFUs [70].

Examples of AFIs for prevention can be found in the physiotherapy domain. One good example is iKneeBraces [113], which helps to prevent knee osteoarthritis. The device evaluates the knee adduction moment using an IMU attached knee brace. AFIs have also been designed for remote health monitoring of older adults, which helps to prevent injuries [58]. Also, prior work has developed an insole based limb load monitoring device with auditory feedback to prevent uneven limb loads for users with walking aids [67].

In the domain of foot orthotics, AFIs can be often found in treatment options. One good example is active ankle-foot orthosis developed for hemiplegic patients to enhance gait parameters by preventing foot drop and toe drag [48]. These devices are instrumented with an insole with FSR sensors identifying impact locations and controls the orthosis accordingly. In addition to the domain of foot orthotics, Walkasins of Rxfunction [116] is another commercially available AFI that improves the balance and walk of patients with sensory neuropathy. Another popular treatment method to improve Parkinson's patients' gait is vibrotactile shoes, which provide rhythmic vibration feedback and rhythmic auditory feedback [65]. These devices act as a treatment for gait freezing of Parkinson's patients.

2.5.6 Context Awareness. AFIs have mainly been used in 2 applications in context-awareness, identifying the user (4 instances) and identifying (one instance) walking terrain. AFIs have used to identify gait parameters for user identification, such as foot motion and pressure [33, 41, 63, 125]. Many of those prior work have used IMU sensors or pressure-sensitive insoles in their AFIs. Matthies et al. presented a capacitive insole-based AFI that can detect walking terrains such as sand, grass, gravel, concrete [64]. Also, it can detect whether the ground surface is wet or electrified. For this application, the insole with capacitive sensors was used to identify different ground couplings demonstrated by different terrains.

2.5.7 Mouse or keyboard like input. In an application where users' hand is occupied with some special task, AFI based input devices have been explored as an alternative to the mouse or keyboard [109,

125]. One such example is for surgeons during operations [126]. A surgeons' hands are usually heavily occupied during operations and controlling information displays by hand is not always possible due to hygiene issues. Hence, several AFIs were developed for surgeons to control information displays by using foot-tapping and scrolling gestures [112, 126]. Some other work also used AFIs with Gaze input as an additional input for gaze typing [94, 95]. Another work explored the viability of foot-tapping gestures using a sock-based AFI in a standing position to control smart devices [32].

3 DISCUSSION

3.1 Challenges and Opportunities

The AFIs that we discussed under our classification have answered several challenging problems in the interaction and human augmentation domain. In table 1, we list those challenges and state how AFIs can provide a unique opportunity in answering those challenges.

3.2 Design Considerations for Future Research

Designing useful AFIs primarily depends on how the creators utilize their foot senses, anatomy, and actions to enhance human capabilities. By looking at the surveyed body of work and our own exploration in creating AFIs, we compiled a list of considerations in designing a new AFI. Many of these aspects have already appeared in many prior works concerning AFIs that we discussed. We believe these considerations will also expose new opportunities to seek out under-explored territories in the domain.

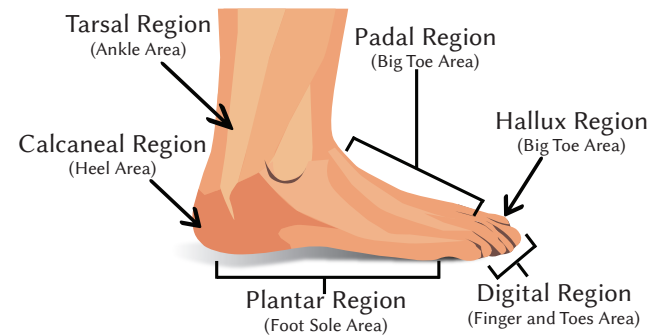


Figure 5: The Foot's main anatomical regions.

Utilizing the anatomy of foot. The foot mainly has 6 anatomical regions: Tarsal (Ankle area), Calcaneal (heel area), Pedal (Foot area), Plantar (Foot sole area), Hallux (Big toe area), and Digital (Fingers and toes area) [101]. These regions offer unique opportunities for designing AFIs. For example, the ankle region is the perfect area for designing sensor bands that captures movements. The foot's ankle joint has three types of rotations: dorsiflexion/plantarflexion, abduction/adduction, and inversion/eversion. These movements can be identified using bend sensors and can be easily matched with foot gestures for explicit interactions. Moreover, Hallux and Digital area have been widely used for foot-tapping and scrolling gestures due to those areas' natural gestures. The top of the foot's pedal area has been utilized for mounting sensors and actuators by

Table 1: Challenges in the Interaction where Augmented Foot Interfaces can Offer Opportunities

	Challenge	Opportunity
Immediate	An interface which can operate at any time by the user with minimal effort.	AFIs can easily design for foot wears. Foot wear is often readily available for all of us. Hence, there is an opportunity to design AFIs available immediately anywhere anytime.
Discreet	An unobtrusive interface that enables privacy for the user and can be used inconspicuously	AFIs can be easily hidden inside Foot wears. Form factors like smart insole or a sock. Although unobtrusive input for AFIs is widely explored unobtrusive silent feedback has yet to be explored.
Bio-mechanical parameter sensing	An interface which can sense important bio-mechanical parameters of the body	Parameters like CoP, limb length equality, impact forces plays a major role in keeping healthy posture and prevent long-term injuries. AFIs can directly help to sense those vital bio-mechanical parameters.
Non-Verbal Behaviour Sensing	An interface which can identify non-verbal behaviors to infer mental status	Anecdotal reports and prior research in behaviour science suggest that involuntary foot gestures may reveal inner mental states of a person. An AFI could be used to sense these gestures.
Subtle and efficient feedback	An interface that can provide feed back in low magnitude, low power, and high resolution.	Feet are densely populated with sensory extremities like free nerve ends, mechanoreceptors and nociceptor. Hence, it is easy to design subtle lowpower, efficient feedback such as tactile.
Multipurpose	An interface with a broad range of utility, application, and reusability	Shoes are something we daily wear. Designing AFIs around foot and shoe will ensure it can be used in a range of applications such as sensing physiological / behaviour parameters and context awareness.

using form-factors such as socks and shoes. The most interesting area is the foot's sole, consisting of highly sensitive nerve receptors such as free nerve receptors, mechanoreceptors, nociceptors, and Meissner's corpuscle. Moreover, these are located closer to the glabrous (hairless) skin surface at the foot's sole, making this area notably more sensitive than others [4, 29, 47]. However, this quality has not been widely utilized in the AFI domain. Only a few works have designed AFIs for navigation and communication utilizing the rich sensitivity of the foot's sole to sense feedback [39, 66]. Also, the foot's high sensitivity to temperature has only been used in trivial applications such as warming the foot [19]. VR-related applications have already used the foot sole's high sensitivity to recreate sensations such as tickling and sensations of different

ground surfaces [54]. However, AFIs for creating sensations such as tickling for daily applications are under-explored.

From sensing physical behavior to mental behavior. Many AFI applications track parameters such as translation and rotation of the foot, foot pressure, and bending angles. From these parameters, many infer physical behaviors, such as activity, posture, anomalies, fatigue. However, anecdotal evidence suggests that foot gestures may have information about the mental status of a person [71, 77]. For example, Joe Navarro's book "what everybody is saying" states the foot as the part of the body most likely to reveal the person's true intention [77]. In a study with students learning foreign languages, researchers found that anxious students continuously tapped their

feet, bounced, and jiggled. In contrast, non-anxious students demonstrated crossed and uncrossed their legs a few times [35]. Considering these anecdotal evidences, and other research on behaviour science, using foot interfaces to infer mental behaviors was first suggested by Velloso et al. [114]. Later, StressFoot [22] conducted a study exploring an AFI capturing foot pressure and motion to detect stress during sedentary tasks. The work showed an F1 score of 85% in a lab setting and substantial robustness in an in-the-wild study. Although StressFoot provides evidence of using foot motion and pressure to detect stress, this area is still not completely explored. For example, the possibility of sensing stress in other postures like standing or walking is still unknown. The sensing accuracy may substantially be improved by using different sensors and more sophisticated AI algorithms. The easy stress tracking provided by mounting a simple IMU sensor on foot may open up many easier stress intervention methods. Also, identifying other emotions such as happiness, sadness, and anger using foot motion and posture characteristics is still open. These gaps pose an immense opportunity to researchers in affective computing to understand features and designers to create AFIs to track those features

Implicit over Explicit. As we mentioned earlier, the first AFI was designed for explicit interaction, as Shannon et al. reported for communication inconspicuously [112]. However, the recent prior work we surveyed showed that many AFIs are based on implicit interactions (81 instances) for applications like gait analysis to detect anomalies [15, 79], posture detection [21, 22], activity recognition [58, 122], and identifying pressure and temperature for medical applications [53, 91, 119, 128]. The AFIs facilitate explicit interactions, primarily provide solutions when the hands are busy. For example, surgeon shoes provide input capabilities during surgical operations [126]. Many work blend AFIs with other inputs such as gaze to improve efficiency [94]. These examples may imply that AFIs are easy to design for implicit interactions compared to explicit interactions. Also, they can be used in a wide range of applications that require implicit interactions. This could be mainly because the foot is primarily used for walking and is engaging with the external world primarily as our main tool of locomotion. Also, naturally, we do not use explicit foot gestures often, as it is considered disrespectful in many cultures. However, many gestures may unconsciously reveal our inner intentions identified in anecdotal evidence that we mentioned before. Prior work such as StressFoot [23] was already designed to understand those behaviors implicitly. Hence in the future, AFI designers should consider that they may be better for implicit interactions. Explicit interactions may come in handy whenever hands are not free or need to perform interactions discreetly.

AFI as a discreet companion. Smartwatches and smart phones already act as our companion device for non-discreet interactions, such as taking photos and other hand gestures. In contrast to hand-held devices or hand-worn devices, foot-worn devices capture less social attention due to its location. Since AFIs can be used discreetly, as already shown in prior work [112]. AFIs have offered unique discreet form factors such as smart insoles, smart socks, and hidden sensors in smart shoes. Also, knee braces and ankle braces are another form factor that may provide discreetness in a location like a gym to prevent unnecessary social attention [7, 81]. We also

always wear shoes, especially when we work or engage in high-risk activities or operate in high-risk locations such as factories, workshops, or while traveling in unknown terrains. This quality can be used to design AFIs for discreet computing. Some commercially available smart shoes, smart insoles, and smart socks already capture users' activity level, fatigue level, parameters like stride length, and frequency while jogging. In research, work like GymSoles [22] proposes how smart insoles can be used as an unobtrusive technology to improve exercise postures. Also, StressFoot [23] shows how AFIs can be used for detecting stress. Combining all these aspects, a shoe-based AFI can be used as a discreet companion, which supports many implicit and explicit interactions.

Designing Comfortable and Usable AFIs. Designing for comfort and aesthetically appealing devices is important for body-worn interfaces. Several commercially available AFIs have already designed comfortable and aesthetically appealing smart shoes [85, 88], insoles [119, 120], and ankle bands [34, 92, 127]. However, designing these for several sizes may be challenging, especially in applications where the spatial resolution of sensor placement is important. For example, developing pressure-sensitive insoles for different people would be difficult as the foot and size of the foot arch have large individual differences. Hence, it is important to find sensors and actuators which are comfortable but not limited by individual differences. IMU sensor tags can be considered a good solution. A simple IMU sensor tag can mount on a shoe. The form factor of the sensor tag can be designed in an aesthetically appealing way. However, a parameter like CoP is difficult to measure using IMU sensors tags. A sensor that can infer CoP by looking at the foot's skin micro-movements may be a viable solution. Google soli may be a possible sensor to capture those micro-movements to infer unknown parameters such as CoP.

AFIs in Assistive Tech. AFIs have many opportunities in assistive technology. Many people suffer lower extremity injuries in their lifetime, and a AFI can help prevent those injuries. For example, a AFI, which provides information about CoP, has been used to reduce injuries while exercising by improving exercise postures [21, 22]. A similar AFI can be used to identify limb imbalances, which is known as acute limb length discrepancy, and may occur due to bad posture. Also, devices like passive haptic rehabilitation gloves have been explored in stroke patients' upper extremity rehabilitation. A similar method can be used for lower extremity rehabilitation. A vibrotactile insole that can provide therapeutic vibration patterns may work as a passive haptic rehabilitation device for the lower extremity. To expand the application possibilities of AFIs in assistive augmentation, it is important to see different application domains other than the intended purpose of its technology. Improving the balance of patients with neuropathy by providing vibration feedback to the ankle is one example of using a different technique, such as sensory substitution AFIs.

4 CONCLUSION

The intersection of Assistive Augmentation with foot interfaces has given rise to a new category of foot interfaces, which we call AFIs (AFIs). We define AFIs as self-contained foot-worn mobile interfaces that extend human abilities with sensing or output capabilities.

Over the past years, interest in AFIs has gained momentum in both academia and the commercial wearable device domain. While prior art surveyed the general research avenue of foot interfaces, we unraveled the notion of assistive augmentation for foot interfaces by providing a more zoomed in view. We investigated common themes of existing work on AFIs and sweet spots for future AFI research. We did so by contributing a survey of the body of work on AFIs. We classified over 100 academic and commercial work to discuss the integration of such devices, interaction schemes, and application domains. Finally, we contribute a set of design considerations for future research of AFIs inspired by the surveyed work.

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